Yash Raj Singh

Senior Software Engineer | React.js | TypeScript | Frontend Architecture

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SUMMARY

Senior Software Engineer with 3+ years building scalable, enterprise-grade web applications in React, JavaScript (ES6+), and micro-frontend architectures. Core maintainer of a shared component platform (1,400+ commits) powering 20+ federated applications used by 40+ enterprise retail clients. Key contributor to the platform's shared API layer, real-time collaboration features, and high-performance data-grid system. Promoted to Senior Engineer within 2.5 years for consistently delivering high-impact features end-to-end - from architecture and performance optimization to production support. Strong in frontend architecture, state management, Module Federation, performance tuning, and developer experience, with hands-on exposure to CI/CD, containerization, and cloud deployment.

EXPERIENCE

Impact Analytics, Bengaluru, India

Senior Software Engineer (Jan 2026 – Present) • Software Engineer (Aug 2023 - Dec 2025) • Software Engineer Trainee (Nov 2022 - Jul 2023)

- Contributed to the platform's shared API layer (Axios interceptors), implementing **HMAC-SHA256 request/response integrity signing** with JWT and CSRF handling to add tamper-resistance across API calls in 20+ federated applications.
- Architected and shipped a **Google Sheets–style real-time commenting and chat system** for AG Grid tables, delivering threaded cell-level comments, @- mention notifications, and live multi-user synchronization over WebSockets with cross-tab state propagation via BroadcastChannel.
- Built and maintained the platform's **enterprise data-grid system** on AG Grid Enterprise - skeleton loaders for client- and server-side tables, custom comparators, dynamic column formatting/value getters, cell wrapping, and debounced auto-sizing - powering data-heavy screens across the product suite.
- Led incremental extraction of the ticketing, release-notes, invalidate-cache, and home modules out of the shared core into the container application, **decoupling the shared library**, shrinking its footprint, and improving maintainability and independent deployability of micro-frontends.
- Designed a **conflict-free, configurable keyboard-shortcut framework** using a custom React hook (keydown/keyup lifecycle, held-key tracking via refs, system-shortcut override, input-field suppression), enabling customizable key bindings and faster navigation for power users.
- Reduced frontend build times by **~65%** by migrating from Babel Loader to esbuild-loader and improved load performance through Module Federation **code splitting** (vendor/MUI/AG Grid/Highcharts chunks), tree shaking, and lazy, route-level dynamic imports.
- Engineered a **cascading cross-filter and saved-filter system** - resolving auto-apply race conditions, correcting saved-filter reapplication, and adding row/column-level data access authorization - for consistent, permission-aware filtering across screens.
- Rebuilt the notifications experience (v3) as a config-driven, tenant-aware module with typed action handlers and real-time updates, and integrated **PostHog** for product analytics, session recording, and observability.
- Co-developed a **metadata-driven rendering engine** that generates fully functional screens from backend JSON - recursively mapping configuration to React components via dynamic imports and an action DSL - reducing manual application setup time by **~70%**.
- Optimized core layout, sidebar, and home components and eliminated duplicate API calls, improving application responsiveness; resolved critical production issues that supported onboarding of **8+ enterprise clients** post-migration.
- Built reusable UI components and a shared **design-system layer** (MUI, AG Grid, Highcharts, MUI X Charts) - charts, forms, date pickers, and cell renderers - reducing feature development effort by **~20%** and improving design consistency.

EDUCATION

Bachelor of Computer Applications

Dec 2020 - Jul 2023

Vivekananda Institute of Professional Studies, GGSIPU - New Delhi

CGPA: 9.42/10

PROJECTS

CoonConvert - Browser-Based Document Toolkit

- Built a client-side document-processing tool (PDF to image conversion, merge, split, compress) integrating multiple PDF libraries behind a unified, responsive UI, with all processing performed in-browser.

ACE Website & Admin Dashboard - Micro-Frontend

- Co-developed a micro-frontend platform with an admin dashboard managing dynamic content (events, blogs, members, galleries, notices), contributing to the MFE architecture, reusable components, and multiple dashboard modules.

SKILLS

Languages	JavaScript (ES6+), TypeScript, HTML5, CSS3/SCSS
Frontend	React.js, Redux, Redux Toolkit, Redux Thunk, React Router, Material UI (MUI v5), AG Grid Enterprise, Highcharts, Styled Components, Responsive Design, Accessibility (a11y)
Architecture	Micro-Frontend Architecture, Webpack Module Federation, Component-Driven Development, Server-Driven / Metadata-Driven UI, Design Systems, Clean Architecture, Design Patterns, System Design
Performance	Code Splitting, Lazy Loading, Tree Shaking, Memoization, Bundle Optimization, Client-Side Caching, Web Vitals
Real-Time / Web	WebSockets, Server-Sent Events (SSE), BroadcastChannel, Cross-Tab Synchronization, REST APIs
Build & DevOps	Git, Github, Webpack, esbuild, Babel, CI/CD (Bitbucket Pipelines)
Testing	Jest, React Testing Library
Tools & Practice	Git, Bitbucket, GitHub, Jira, Confluence, Agile/Scrum, PostHog (analytics & observability), AI-assisted development (Cursor, Windsurf, Claude)

AWARDS

- Spot Award for Excellence -Q4 2023, Impact Analytics (Feb 2024)
- 2nd Place - HackJMI Hackathon (Feb 2023)